

SILAS KWABLA GAH

PhD Candidate in Computer Vision · Foundation Models, Spatial AI & Multimodal Reasoning

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Research Profile

Final-year PhD candidate in Computer Vision and Machine Learning at the University of Ghana (thesis defence scheduled for October 2026). Specializes in **inference-time foundation-model composition**, **open-vocabulary spatial perception**, **multimodal vision-language reasoning**, and **probabilistic calibration under extreme data scarcity** ($\Delta\theta = 0$). Experienced in processing multi-sensor visual feeds, 2D-to-3D geometric lifting, camera projection geometry, and self-supervised reinforcement learning process rewards (GRPO on vLLM). Directly aligned with the **College of Computing (COLCOM) at UM6P**: developing scalable, low-supervision AI solutions for **remote sensing**, **drone/satellite agricultural monitoring**, open-vocabulary canopy segmentation, and multimodal knowledge extraction, scaled across the **TOUBKAL** supercomputing infrastructure.

Education

PhD in Computer Vision and Machine Learning

University of Ghana, Legon

Expected defence: October 2026

Supervisor: Prof. Ebenezer Owusu

Thesis: *Visual Perception under Low Supervision: Inference-Time Coordination of Frozen Foundation Model Representations*

Investigates open-vocabulary spatial segmentation under extreme label scarcity, multi-view geometric consensus, mitigating premature decision collapse, and reliability-calibrated inference across heterogeneous visual backbones.

MPhil in Computer Science

University of Ghana, Legon

2016

Accra, Ghana

BSc in Information and Communications Technology

Ghana Institute of Management and Public Administration (GIMPA)

Second Class Honours (Upper Division) · 2012

Accra, Ghana

HND in Electrical & Electronic Engineering

Accra Technical University

2009

Accra, Ghana

Doctoral Research Highlights & Systems Rigor

Open-Vocabulary Spatial Perception & Geometric Consensus (ACCV 2026, WACV 2027, ICLR 2027)

- Developed training-free open-vocabulary spatial segmentation frameworks reconciling dense vision-language embeddings with 2D promptable segmenters (SAM 2.1) via camera projective geometry (K, T), ray casting, and spatial consensus. Directly transferable to multi-spectral drone and satellite agricultural remote sensing.
- Evaluated across large-scale benchmarks ($\sim 50M$ spatial points, 312 scenes), achieving state-of-the-art harmonic mean (34.87 novel mIoU, +6.40 points improvement) with paired bootstrap 95% confidence intervals [5.24, 7.64].
- Formalized the mechanism of **premature semantic collapse** under hard argmax selection; proved that retaining compact top- k candidate distributions preserves rare target categories that standard pipelines discard (formalized in an **ICLR 2027** submission).
- Engineered high-throughput evaluation pipelines with float16 caching and chunked processing to evaluate dense spatial distributions on high-performance GPU clusters.

Multi-View Physical Consistency as Intrinsic Self-Supervision (SACAIR 2026 Submission)

- Formulated physical consistency across independent viewpoints as an intrinsic, label-free process reward for multimodal reasoning models (**Qwen2.5-VL-7B**) using online **GRPO/VERL** on **vLLM**.
- Mitigated reward-hacking and area-bias specification gaming ($\rho > 0.72 \rightarrow 0.334$) without human-labeled reward annotations; framework directly extends to grounding LLM/VLM reasoning on multi-angle drone agricultural imagery.

Reliability Calibration & Uncertainty Quantification (RCASA) (Target: IEEE TPAMI In Prep)

- Formulated **RCASA**, a principled probabilistic framework uniting Platt confidence calibration with Bernoulli uncertainty estimation to decouple classifier confidence from contextual reliability.
- Proved class-wise bounded updates and probability simplex preservation, providing rigorous trust guarantees for detecting rare agricultural anomalies, plant pathologies, and out-of-distribution environmental conditions.

Selected Publications and Manuscripts

- **Beyond Argmax: A Mechanistic Study of Semantic Retention in Frozen Foundation-Model Composition for Generalized Few-Shot 3D Segmentation.**
Intl. Conference on Learning Representations (ICLR 2027 Submission; Preprint Available). First Author.
- **Training-Free Open-Vocabulary 3D Point-Cloud Segmentation on the Generalized Few-Shot Benchmark.**
Asian Conference on Computer Vision (ACCV 2026, Paper #482; Accepted / LNCS). First Author.
- **Training-Free Generalised Few-Shot Segmentation through Open-Vocabulary Semantic Arbitration.**
IEEE/CVF Winter Conference on Applications of Computer Vision (WACV 2027, Paper #167). First Author.
- **Cross-View Agreement as a Label-Free Process Reward for Multimodal Reasoning.**
Southern African Conference for AI Research (SACAIR 2026 Submission; Revised). First Author.
- **Reliability-Calibrated Adaptive Semantic Arbitration of Frozen Foundation Model Representations.**
Target: IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI) — In Preparation. First Author.
- **Transfer Learning-Based Encoder-Decoder Model for Skin Lesion Segmentation.**
Communications in Computer and Information Science (CCIS), Vol. 1841, pp. 117–128, Springer, 2023. Co-Author.

Technical Skills

Programming	Python, C++, Java, Shell scripting
Deep Learning & RL	PyTorch, Hugging Face, Transformers, vLLM, LoRA/QLoRA, SFT, GRPO, VERL, Accelerate
Computer Vision	Open-vocabulary segmentation, SAM 2.1, CLIP, multi-view geometry (K, T), ray casting, 3D point clouds, spatial scene representations, remote sensing imagery
Multimodal AI Calibration & Uncertainty	Vision-Language Models (Qwen2.5-VL), multimodal grounding, label-free process rewards Platt scaling, Expected Calibration Error (ECE), Brier score, paired bootstrap CIs, risk control
Systems & HPC	Slurm, CUDA fundamentals, float16/bfloat16 distributed pipelines, multi-GPU scaling (TOUBKAL-ready)

Teaching, Professional Experience & Academic Service

Part-time Lecturer	Accra Technical University (2022 – Present)
• Lecturing in machine learning, algorithms, and data structures; undergraduate student project supervision.	
Founder & Lead Architect	EED Soft Consult (2018 – Present)
• Architected and deployed enterprise software, distributed services, and cloud pipelines in Ghana.	
Academic Service	Reviewer, SACAIR 2026 Program Committee

References

Prof. Ebenezer Owusu PhD Supervisor Dept. of Computer Science University of Ghana, Legon Email: eowusu@ug.edu.gh	Prof. Jamal-Deen Abdulai Head of Department Dept. of Computer Science University of Ghana, Legon Email: jabdulai@ug.edu.gh	Dr. Justice Kwame Appati Co-Author / Senior Lecturer Dept. of Computer Science University of Ghana, Legon Email: jkappati@ug.edu.gh
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